

KARTIK SINGLA

Senior Frontend Engineer | React • TypeScript • Next.js • Frontend Architecture

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PROFESSIONAL SUMMARY

Senior Frontend Engineer with 5+ years building and scaling production frontend systems. Led the shift from a monolithic frontend to a 6-app Turborepo micro-frontend architecture for a healthcare platform, and built a shared design system adopted across 8+ modules that cut new-feature development time by approximately 30%. Specializes in frontend architecture, React/Next.js ecosystems, performance optimization, and integrating AI-assisted engineering workflows (Claude Code) into team delivery.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3

Frontend: React.js, Next.js, React Hooks, Responsive Web Design, Component-Driven Development

Frontend Architecture: Micro Frontends, Turborepo, Monorepo Architecture (pnpm workspaces), Design Systems, Shared UI Libraries, Reusable Component Architecture

State & Data: TanStack Query, Zustand, Redux, REST APIs, GraphQL, OpenAPI-Generated Types, API Integration

UI & Styling: Tailwind CSS, shadcn/ui, Ant Design, Material UI, Storybook

Build & Developer Tooling: Vite, Turborepo, pnpm, ESLint, Prettier, Git

Testing & Quality: Vitest, Jest, Unit Testing, Component Testing, Code Review, Type Safety

Performance & Web Engineering: Web Performance Optimization, Core Web Vitals, Code Splitting, Lazy Loading, Caching, Accessibility (WCAG 2.1 AA)

Realtime & Integrations: Real-Time Event Architecture, WebSockets, Third-Party SDK Integration, Video/Telehealth Integrations

AI-Assisted Engineering: Claude Code (agentic coding workflows), AI-based code review

Engineering Practices: SOLID Principles, Clean Code, DRY, KISS, Agile/Scrum, CI/CD, Technical Documentation, Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Addval Solutions — Mohali, India | July 2025 – Present

- Own frontend architecture for the Kulcare healthcare platform — patient management, appointments, billing, and telehealth — across a Turborepo micro-frontend system spanning 6 independently deployable applications.
- Led the shift from a monolithic frontend to Turborepo as platform scope grew across multiple clinical workflows, enabling independent team releases without blocking shared feature work.
- Introduced AI-assisted development practices — Claude Code for agentic implementation, AI-based code review — into the team's workflow, adopted across a 5-engineer team, to speed delivery and surface issues earlier.
- Built a shared design system and component library adopted across 8+ application modules, replacing duplicated implementations and cutting new-feature development time by approximately 30%.
- Owned telehealth/video-consultation workflows end-to-end and designed a real-time event architecture to keep appointment, queue, and notification data consistent across patient-facing and CMS applications.
- Built frontend flows for ABDM/ABHA (India's national digital health mission) integration — including HIP linking, ABHA authentication, QR-based patient identification, and consent workflows — enabling compliance with government health-data interoperability standards.
- Raised accessibility standards across core clinical workflows — keyboard navigation, screen-reader support, contrast — bringing key flows to WCAG 2.1 AA.
- Closed the type-safety gap between backend and frontend by auto-generating API types from the OpenAPI schema, removing a recurring source of integration bugs, and diagnosed rendering/load-time bottlenecks that cut page-load time by approximately 25%.
- Set technical direction for a 5-engineer frontend team — code review standards, TypeScript conventions, and architecture decisions.

Senior Software Engineer

Wits Innovation Lab — Mohali, India | July 2024 – June 2025

- Built and extended a multi-role ONDC-based e-commerce marketplace (React, Saleor) serving buyer, seller, and admin workflows across 100+ sellers and 5,000+ SKUs.
- Replaced legacy REST integrations with optimized GraphQL as catalog and order volume grew, cutting average data-fetch time by approximately 40%.
- Designed configurable commerce workflows that cut content-update turnaround from approximately 2 days to under 2 hours, reducing engineering dependency for routine catalog changes, while partnering with product, backend, and QA teams on maintainable, scalable frontend solutions.

Software Developer

Seasia Infotech — Mohali, India | August 2021 – July 2024

- Built Zanran, a document-intelligence platform, from the ground up — search, extraction, visualization, and analytics workflows for 10,000+ enterprise documents.
- Designed advanced search and filtering that cut average document-lookup time from approximately 5 minutes to under 30 seconds for 200+ analysts and auditors.
- Developed interactive PDF data visualization and extraction workflows, translating complex document-processing requirements into an intuitive review experience.

EDUCATION

B.Tech, Computer Science

CGC Technical Campus — Faculty of Engineering | 2021